

Gas laws

THE GAS LAWS STATE HOW GASES RESPOND TO CHANGE.

The three laws relate the movements of molecules in a gas to its volume, pressure, and temperature, and state how each measure responds when the others change. Each gas law is named after its discoverer.

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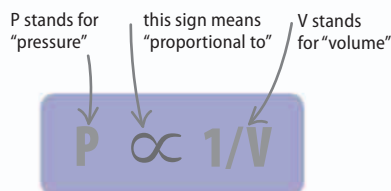
◀ 99 Gases

Pressure 141 ▶

Pressure 184–185 ▶

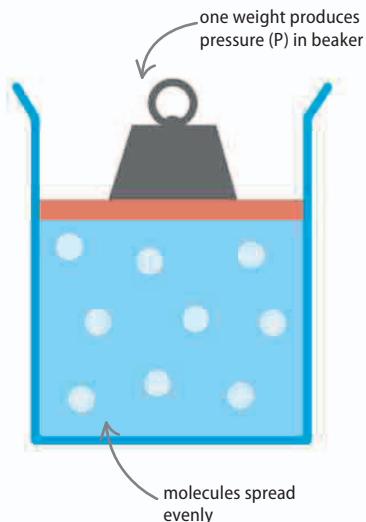
Boyle's law

This law is named after Robert Boyle, who lived in Britain and Ireland in the 17th century and was one of the world's first chemists. His law states that if the temperature of a gas stays the same, then its volume is inversely proportional to its pressure. In other words, forcing a gas into a smaller volume results in it exerting a higher pressure.



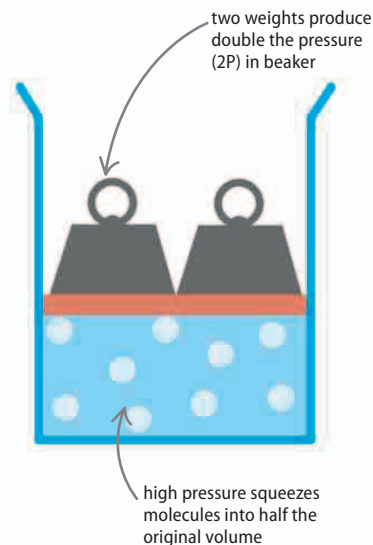
△ Equation for Boyle's law

This equation shows the relationship between a gas's pressure and its volume. Increasing the pressure decreases the volume.



△ Diffusion

The molecules in the gas spread out evenly to fill any container. This is called diffusion and means that molecules tend to move away from places where they are highly concentrated.



△ Pressure

The force exerted on an area (its pressure) is caused by molecules in the gas hitting the inside of the container. Reducing the volume gives the molecules less room to move. They hit the sides more frequently, increasing the pressure.

REAL WORLD

Avogadro's law

There is a fourth gas law, which, although unrelated to the other three, was set out by the Italian Amedeo Avogadro (right) in 1811. It states that equal volumes of all gases at the same temperature and pressure contain the same number of molecules. Therefore a flask of hydrogen can contain the same number of molecules as an identical flask of oxygen, despite weighing a lot (16 times) less.



Robert Boyle was an alchemist and discovered his law while he was searching for a way to turn lead into gold.